

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MIDSEM EXAMINATION)

CLASS: BTECH/IMSc
BRANCH: BT/CHEMICAL/CIVIL/MECH/PIE/FT/PHYSICS

SEMESTER: II
SESSION: SP/2024

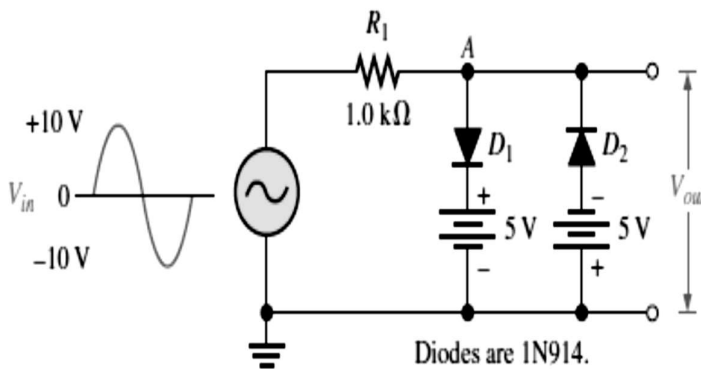
SUBJECT: EC101 BASICS OF ELECTRONICS AND COMMUNICATION ENGINEERING
TIME: 2 HOURS

FULL MARKS: 25

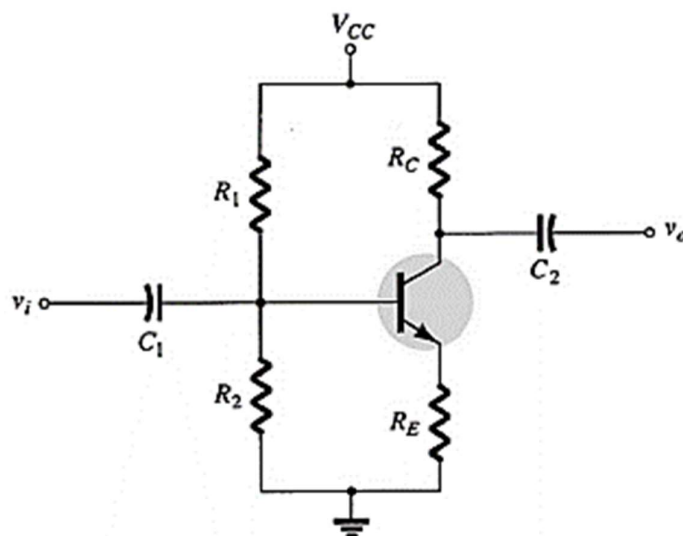
INSTRUCTIONS:

1. The question paper contains 5 questions each of 5 marks and total 25 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates

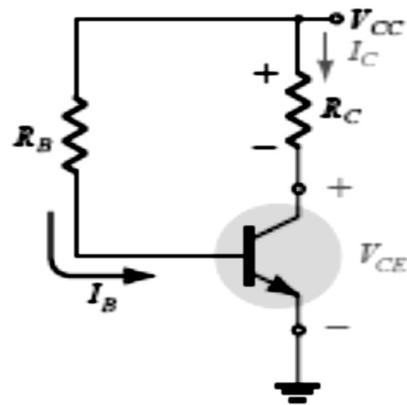
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| Q1(a) Given a diode current of 6 mA, $V_T = 26$ mV, $\eta = 1$, and reverse saturation current $I_s = 1$ nA, find the applied voltage V_D . | 2 | 1 | 2 |
| Q1(b) Explain the working of Centre-tap full wave rectifier circuit and derive its efficiency. | 3 | 1 | 3 |
| Q2(a) Differentiate the Zener breakdown and avalanche breakdown. Draw the V-I characteristic of a Zener diode. | 2 | 1 | 2 |
| Q2(b) Evaluate the output voltage of the following circuit. Assume D1 and D2 are Si and Ge diodes respectively. | 3 | 1 | 3 |



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|--|---|---|---|
| Q3(a) Establish the relation between α , β and γ . | 2 | 2 | 3 |
| Q3(b) In the following Figure, determine the values of I_C , V_{CE} and V_E . The data given are, $R_1 = 62$ kΩ, $R_2 = 9.1$ kΩ, $R_C = 3.9$ kΩ, $R_E = 0.68$ kΩ, $V_{CC} = 16$ V, and $\beta = 100$. | 3 | 2 | 3 |



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|-------|---|---|---|---|
| Q4(a) | Define pinch-off voltage for a JFET. | 2 | 2 | 2 |
| Q4(b) | Explain the working of n-channel JFET. Draw the drain characteristics and transfer characteristics to support your answer. | 3 | 2 | 2 |
| Q5(a) | Differentiate JFET and MOSFET. | 2 | 2 | 2 |
| Q5(b) | Determine I_B , I_C and V_{CE} . Given: $R_B=240\text{ k ohm}$, $R_C=2.2\text{ k ohm}$, $V_{CC}=12\text{ V}$ and $\beta=50$. | 3 | 2 | 3 |



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